UE Projet Informatique
Soutenance

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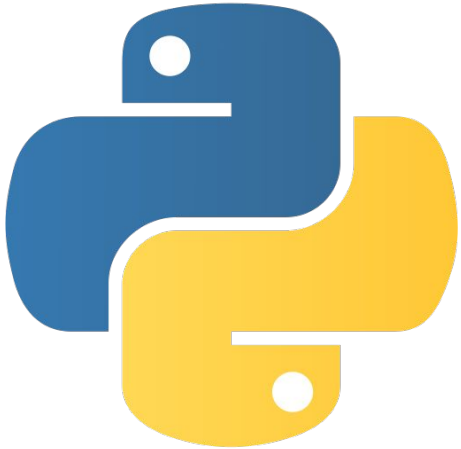
Sommaire



Introduction

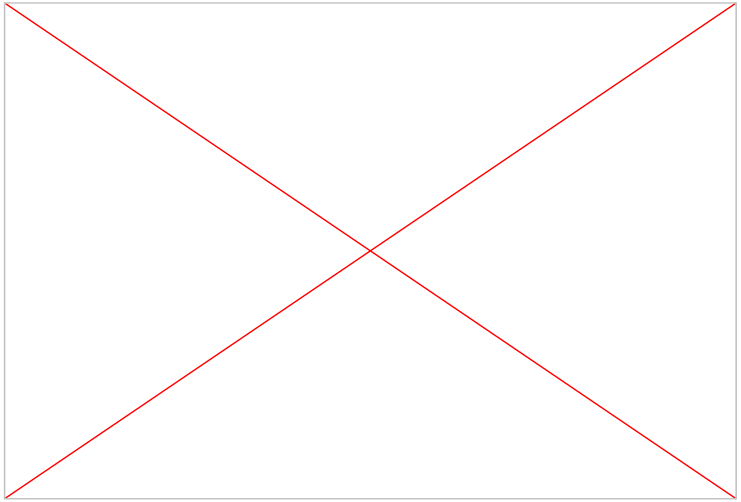
1) Architecture et étapes de développement
2) Diagrammes de classes
3) Chargement des niveaux et persistance en JSON

Conclusion



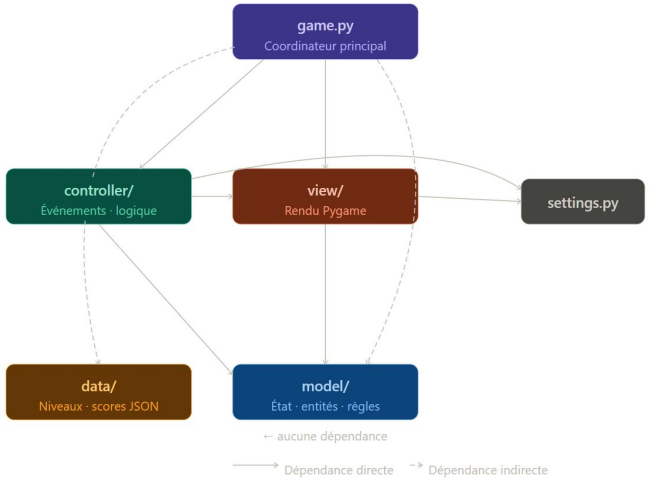
Introduction - Tower Merge Defense

- Développé en python
- Tower Defense
- Mécanique de fusion
- Boosters
- Niveaux



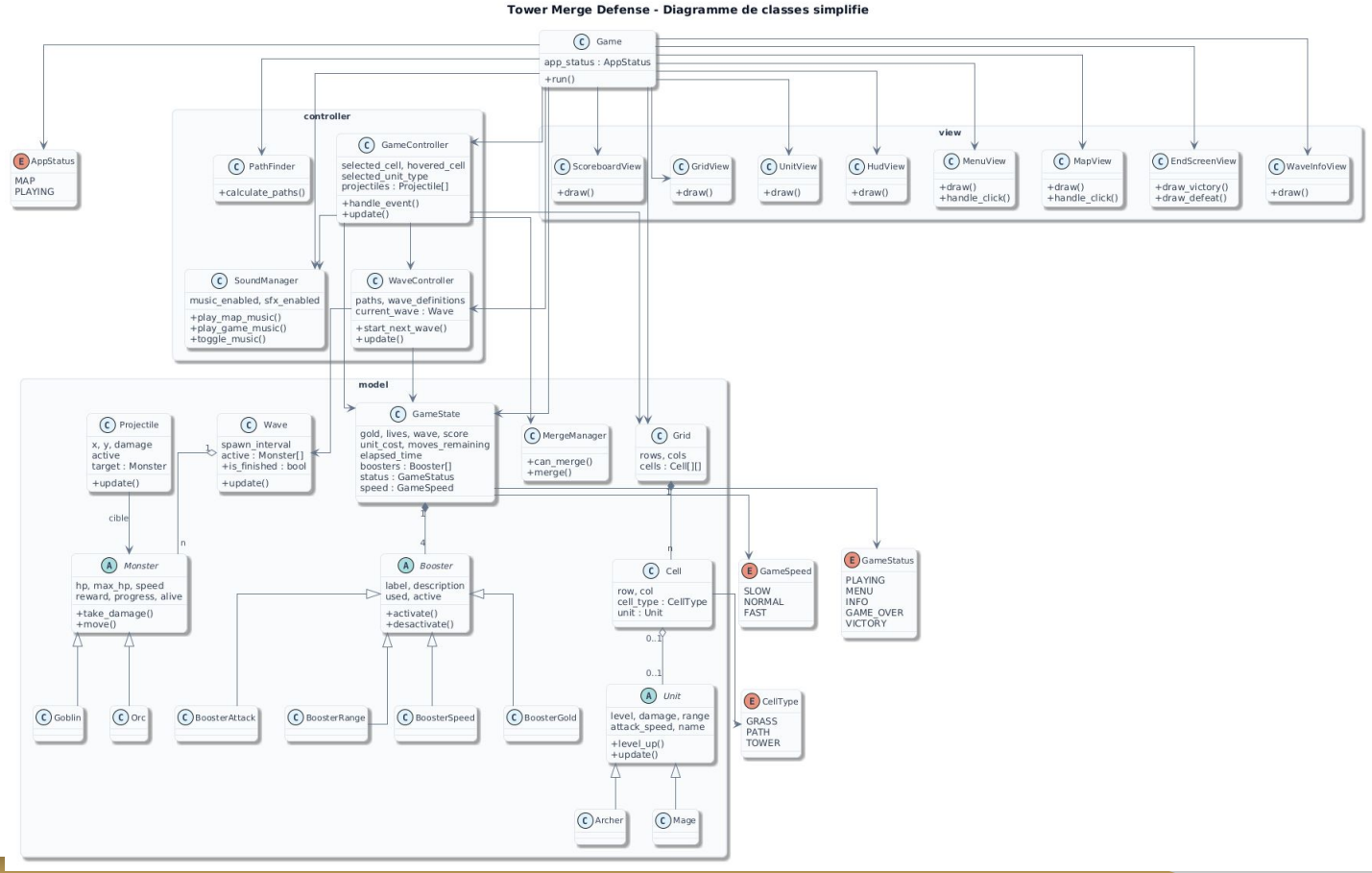
Architecture et étapes de développement

Structure MVC : model-view-controller et classe Game qui fait le lien



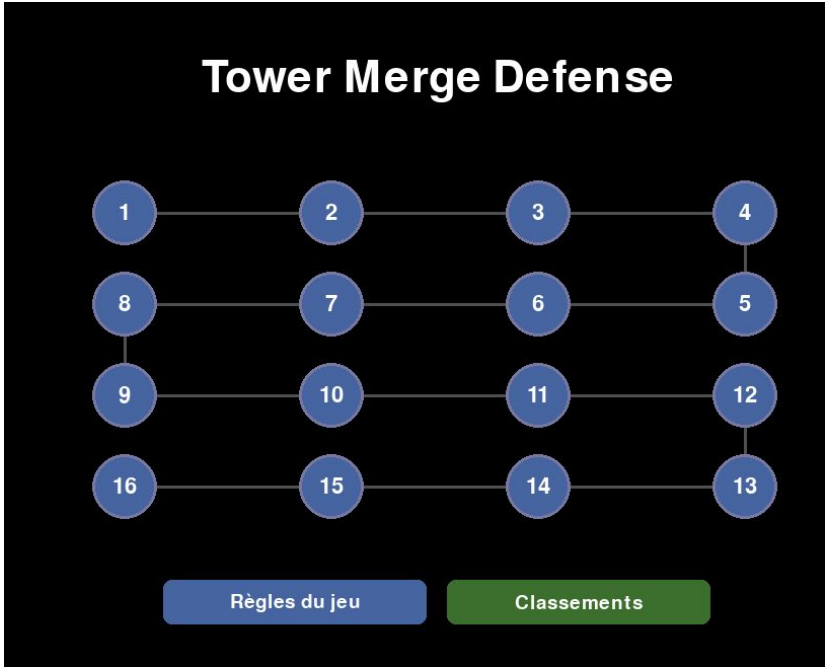
- Grille et calcul de chemin
- Déplacement des monstres saccadé puis fluide
- Unités
- Projectiles
- Fusion
- Vagues
- Interfaces

Diagramme de classes



Chargement des niveaux et persistance des données

```
{
  "id": 1,
  "start": [5, 1],
  "end": [8, 1],
  "layout": [
    [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
    [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
    [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2, 0, 0, 0, 0],
    [0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 0, 0, 0],
    [0, 0, 0, 2, 2, 2, 0, 0, 0, 1, 0, 1, 0, 0, 0],
    [0, 1, 1, 1, 1, 1, 1, 1, 1, 2, 1, 2, 0, 0, 0],
    [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 2, 0, 0],
    [0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 0, 0, 0],
    [0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 0, 0, 0, 0],
    [0, 0, 0, 2, 2, 2, 0, 0, 0, 0, 0, 0, 0, 0, 0]
  ],
  "waves": [
    {
      "spawn_interval": 2.0,
      "monsters": [
        {
          "type": "G",
          "path": 0
        },
        {
          "type": "G",
          "path": 0
        },
```



Conclusion

```
>>> print("Merci de votre attention")
Merci de votre attention
```